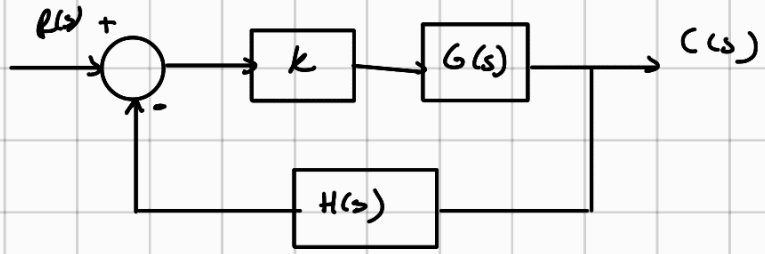
Root locus

All posible root locations of a system

→ For higher order systems the "degree" of polynomial in the charachteristic equation is high!

↳ Finding root becames difficult



accuracy → TF + R(s)

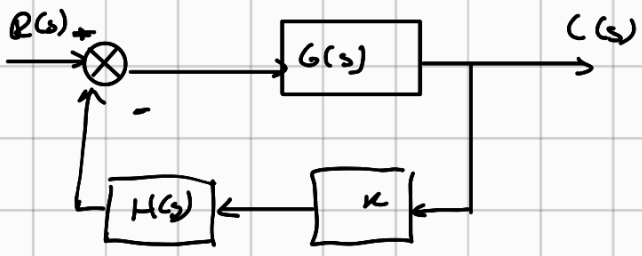
$$\frac{C(s)}{R(s)} = \frac{k\,G(s)}{1 + k\,G(s)\,H(s)}$$

$$G(s) = \frac{N_G(s)}{D_G(s)}$$ (Numerator / Denominator)

$$H(s) = \frac{N_H(s)}{D_H(s)}$$

$1 + k\,G(s)\,H(s) = 0$ → stability

Quality of response (t_s, t_p, M)



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + k\,G(s)H(s)} = \frac{N_G/D_G}{1 + k\frac{N_G}{D_G}\cdot\frac{N_H}{D_H}} = \frac{N_G\,D_H}{D_G\,D_H + k\,N_G\,N_H}$$

CLTF:

- zeros of TF ⇒ "zeroes" of G + "poles" of H
- poles of T.F ⇒ $D_G D_H + k N_G N_H = 0$ ⟶ $k = 0 \Rightarrow D_G D_H = 0$ → poles of G, poles of H

By changing 'k' roots of the characteristic equation change from 'poles' to 'zero' of the 'open loop' T.F.

$k \to \infty \Rightarrow N_G N_H = -\frac{D_G D_H}{k} \to 0$

↳ zeroes of G

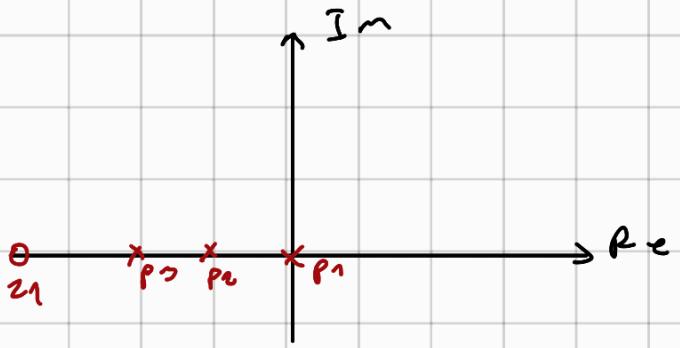
↳ zeroes of H

$$G(s) = \frac{s+5}{s(s+1)(s+2)}$$ → $z_1 = -5$

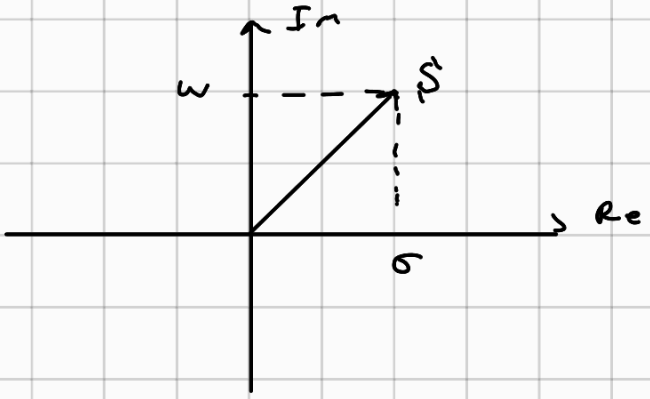
→ $p_1 = 0$

$p_2 = -1$

$p_3 = -2$



Vector representation in complex plane



$\sigma + jw = R e^{j\theta} = R\angle\theta$

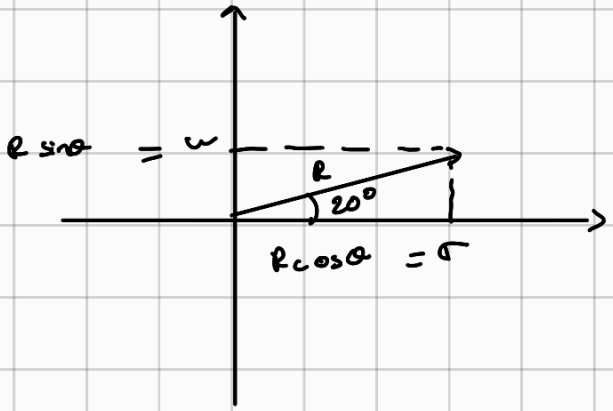
$R = \sqrt{\sigma^2 + w^2}$

$\theta = \tan^{-1}\left(\frac{w}{\sigma}\right)$

origin of complex plane to S'.

$R = 10$

$\theta = 20$



$s = R\cos\theta + jR\sin\theta$

s is complex

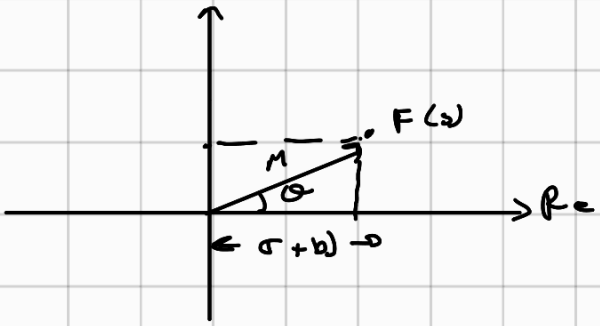
$F(s) = s + b$ for any complex s, we'll end up to as a complex number

$s = \sigma + jw$

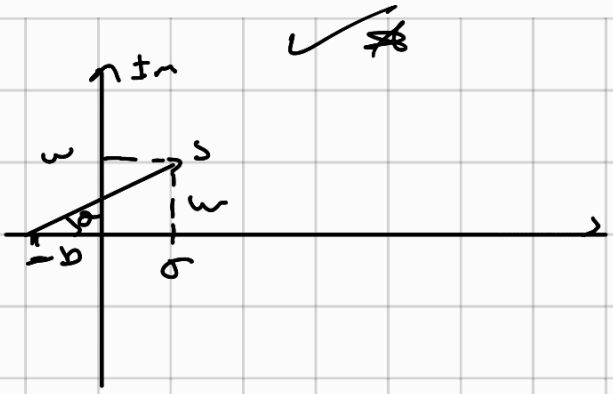
$F(s) = (\sigma + b) + jw = M\angle\theta$

$M = \sqrt{(\sigma+b)^2 + w^2}$

$\theta = \tan^{-1}\left(\frac{w}{\sigma+b}\right)$



Starting from origin of s plane



Starting from origin of F(s)

$$F(s) = \frac{s+a}{s+b}$$ → origin of s+a (numerator)

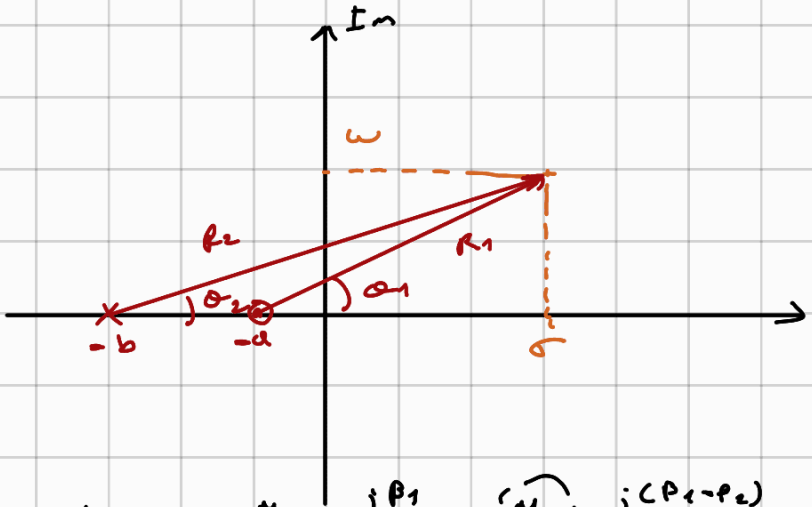
$$\Rightarrow F(s) = \frac{\sigma + jw + a}{\sigma + jw + b} \times \frac{(\sigma+b) - jw}{(\sigma+b) - jw}$$

$s = \sigma + jw$

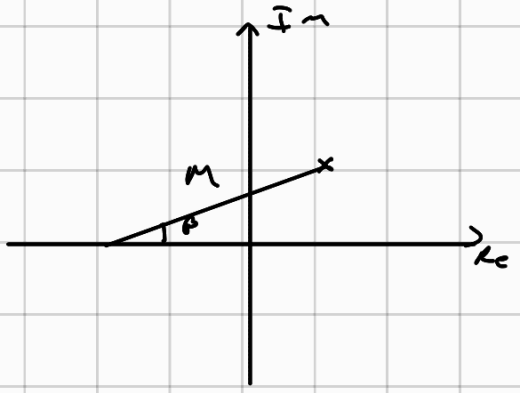
$R_1 e^{j\theta_1}$ $R_2 e^{j\theta_2}$

$$F(s) = \frac{[(\sigma+b)^2 + jw][(\sigma+b) - jw]}{(\sigma+b)^2 + w^2}$$

$$= \frac{R_1}{\alpha} e^{j\theta_1} + \frac{R_2}{\alpha} e^{j\theta_2}$$



$$F(s) = \frac{M_1 e^{j\beta_1}}{M_2 e^{j\beta_2}} = \frac{M_1}{M_2} e^{j(\beta_1 - \beta_2)} = M e^{j\beta}$$



$$F(s) = \frac{\prod_{i=1}^{N} (s+z_i)}{\prod_{i=1}^{m} (s+p_i)} = \frac{(s+z_1)(s+z_2)\cdots}{(s+p_1)(s+p_2)\cdots}$$

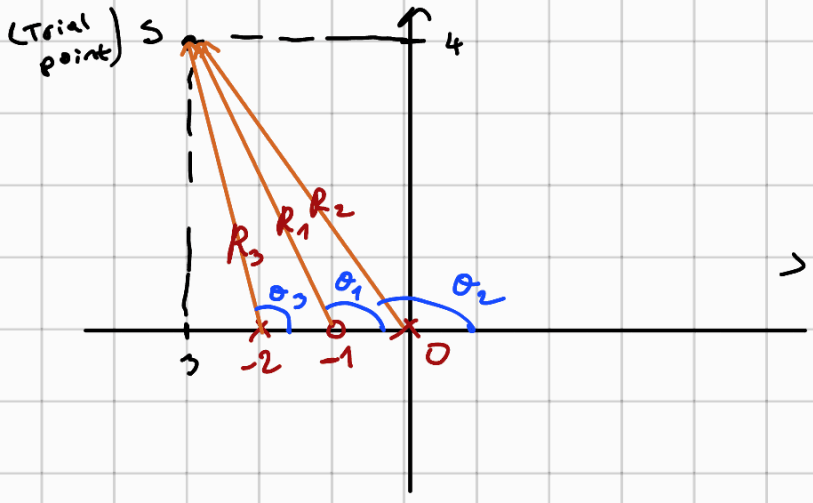
at any s, the function value is a complex number (in general) which has a "magnitude" and a "phase"

$$M = \frac{M_1 \cdot M_2 \cdots\cdots M_N}{M_1' M_2' - \cdots M_m'}$$

θ = Σ Phases of Nomitator − Σ phases of denominator

$$F(s) = \frac{s+1}{(s+2)s}$$

What is the function value at $s = -3 + j4$



For $\vec{R_1}$ { M_1 = magnitude = $\sqrt{2^2+4^2}$
β_1 phase $\pi - \tan^{-1}\left(\frac{4}{2}\right)$

For $\vec{R_2}$ { M_2 mag = $\sqrt{3^2+4^2}$
$\beta_2 = \pi - \tan^{-1}\left(\frac{4}{3}\right)$

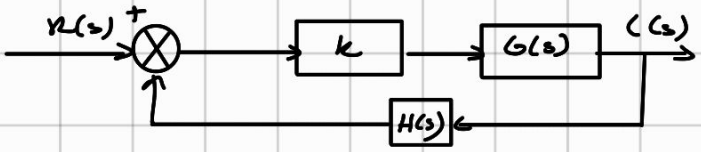
For $\vec{R_3}$ { $M_3 = \sqrt{1^2+4^2}$
$\beta_3 = \pi - \tan^{-1}(4/1)$

$$M = \frac{M_1}{M_2 M_3}$$

$\theta = \underbrace{\beta_1}_{\text{numerator}} - \underbrace{\beta_2 - \beta_3}_{\text{Denominator}}$

4

We need some criteria to check if the trial poinsts (s) on the complex plane can be accepted as root of the characktristic equation



$$\frac{C(s)}{R(s)} = \frac{k\,G(s)}{1 + k\,G(s)\,H(s)}$$

$$1 + k\,G(s)\,H(s) = 0 \Rightarrow G(s)\,H(s) = -\frac{1}{k} + 0j$$

magnitude: $|G(s)\,H(s)| = \left|\frac{1}{k}\right|$

phase: $\angle G(s)\,H(s) = \tan^{-1}\left(\frac{0}{-1/k}\right) = \pm 180(2i+1) \qquad i = 0, 1, \ldots$

↑ because its negative

Any point like 's' which satisfies both conditions would be our point of intereset (root of characteristic eq)

The locus of points statisfying the phase condition constitues the root locus plot of the system.

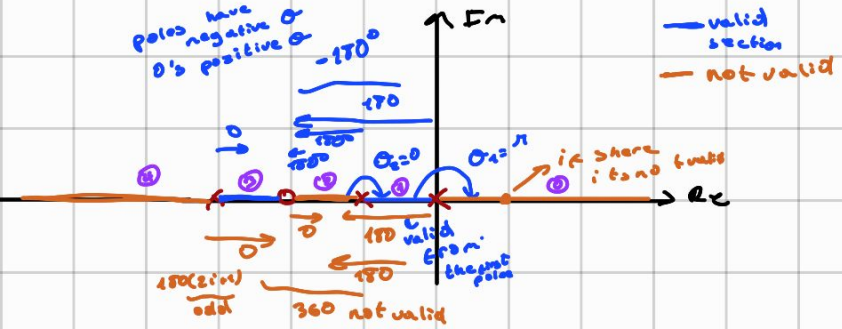
Steps of Drawing

1- Find the OL zeros and poles and show on the s plane

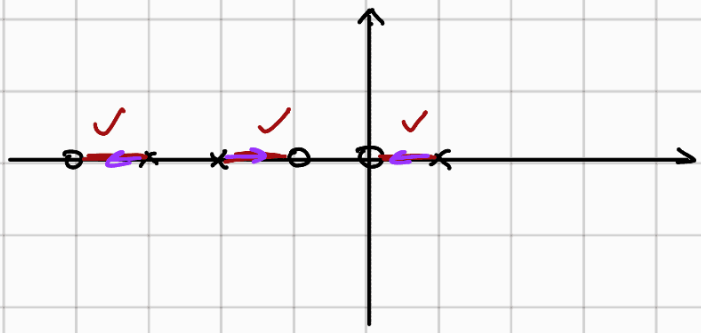
$$1 + k\,G(s)\,H(s) \Rightarrow 1 + k\frac{Q(s)}{P(s)} = 0 \Rightarrow P(s) + kQ(s) = 0$$

- if $k = 0 \Rightarrow P(s) = 0 \Rightarrow$ poles of OLTF — × symbol
- if $k \to \infty \Rightarrow Q(s) = -\frac{P(s)}{k} \Rightarrow Q(s) = 0$ ⇒ zeros of OLTF — o symbol

2- Check the validity of point on 'Re' axis



★ So if the number of zeros/poles on the right hand side of the zone is 'odd', then it pases the phase condition



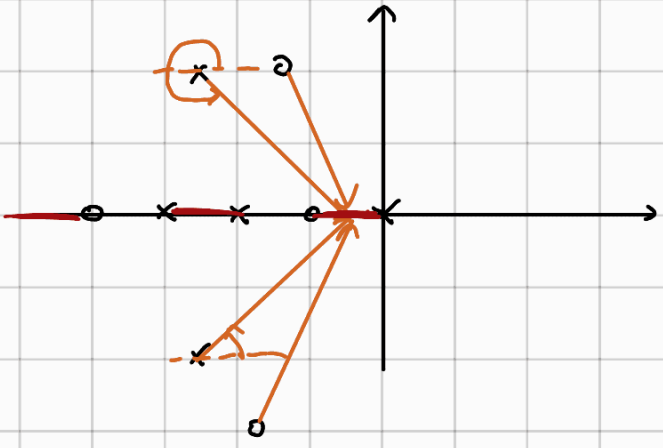
if $k=0 \rightarrow k=\infty$

roots move

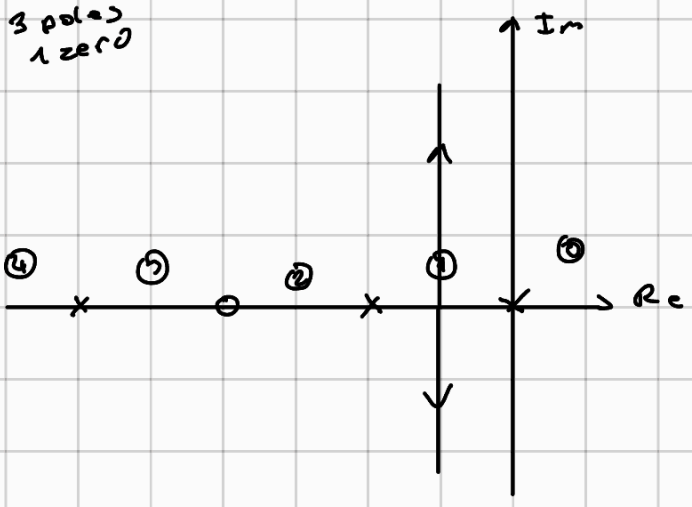
★ poles of CL

start at poles of O.L.

and at zero of O.L.



Complex poles not effect zeroes the acceptible zone at the real axis



All poles have to married with zeros

if they can't they go to infinity

3. Assymptote determine

if the number of poles and zeros are not the same

Then there will will be some asymptotes for the root locus

poles: n

zeros: m

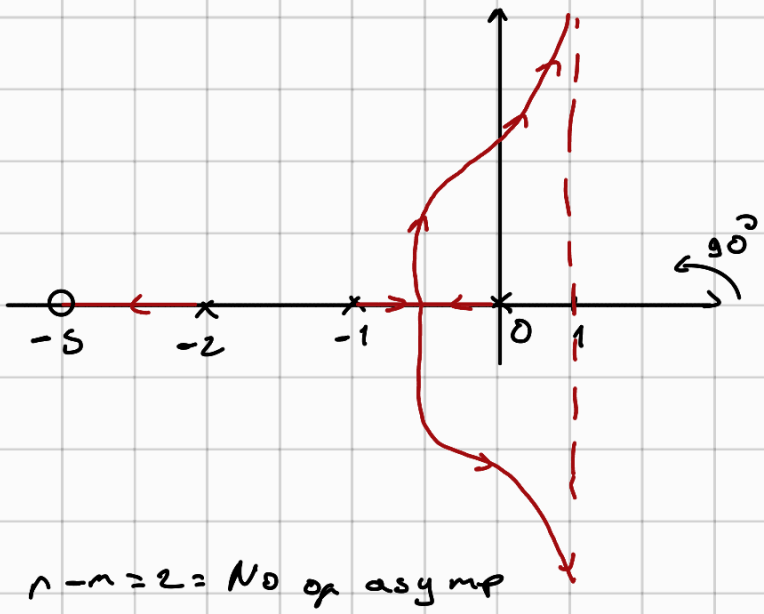
$\Rightarrow n-m$: number of asymptotes

'n' is usally larger than 'm'

★ Number of Branches is equal to 'n'

Angle of asymptotes $\Phi = \frac{(2i+1)\,180}{n-m}$, $i = 0, 1, 2, \ldots, n-m-1$

* Centroide of asymptotes $S' = \frac{\Sigma \text{real part of poles} - \Sigma \text{real part of zeros}}{n-m}$



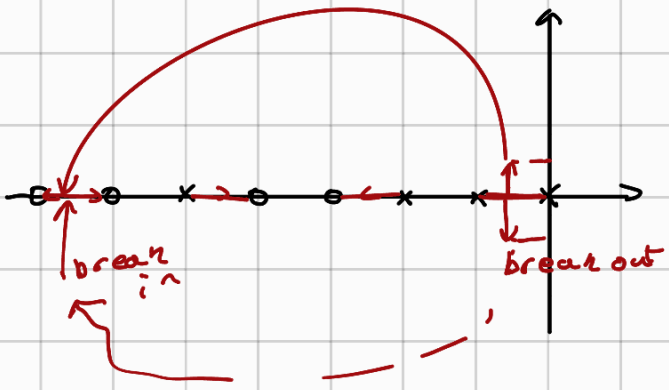
$\Phi = \frac{(2i+1)\,180}{2} = (2i+1)\,90$ → $i=0 \rightarrow \Phi_1 = 90$, $i=1 \rightarrow \Phi_2 = 270$

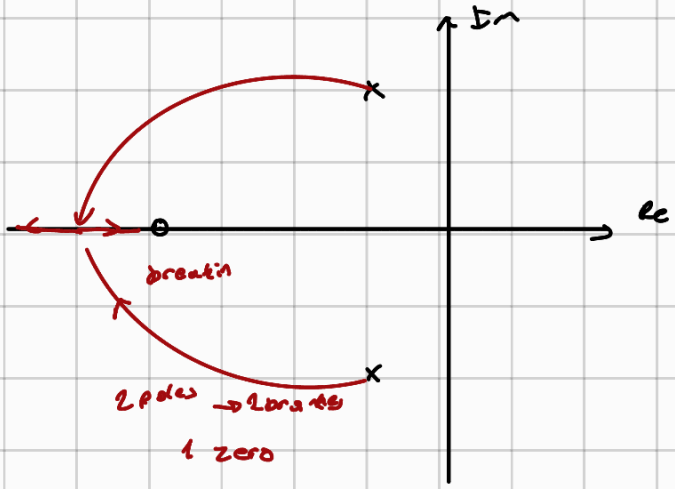
centroid : $\frac{(0-1-2)-(-5)}{2} = +1$

$n - m = 2 =$ No of asymp

Determine the breaking points

* Branches leave (break out) or enter (break in) the real axis at 90°





The charcteristick eq

$1 + k\frac{Q(s)}{P(s)} = 0$

$P(s) + kQ(s) = 0$

$\Rightarrow k = -\frac{P(s)}{Q(s)}$

repeated root occurs at point of break in / out $\rightarrow \frac{dk}{ds} = 0$

↳ $P'Q - Q'P = 0$

All break points are solution of this polynomial But not all solutions are break in/out points

Only root of the given expression are acceptable that give a positive gain!

The other approach is that the break in/out point must be on on the valid zones of the real axis.